

CHAPTER 5

JACOBIAN ANALYSIS OF PARALLEL MANIPULATORS

5.1 INTRODUCTION

This chapter is devoted to the Jacobian analysis of parallel manipulators. The Jacobian analysis of parallel manipulators is a much more difficult problem than that of serial manipulators because there are many links that form a number of closed loops. Study of the instantaneous motion of kinematic chains was pioneered by Waldron (1966), Davies and Primrose (1971), and later Baker (1980). Since then, various methods of analysis have been proposed. Davies (1981) developed a constraint law for a mechanical network that is analogous to Kirchhoff's circulation law. Mohamed et al. (1983) developed a procedure for determination of the instantaneous twists associated with the joints of a limb using velocity vector-loop equations. Mohamed and Duffy (1985) introduced the screw theory, and Sugimoto (1987) applied motor algebra for the Jacobian analysis of parallel manipulators.

An important limitation of a parallel manipulator is that singular configurations may exist within its workspace where the manipulator gains 1 or more degrees of freedom and therefore loses its stiffness completely. This property has attracted the attention of several researchers. Gosselin and Angeles (1990) studied the singularities of closed-loop mechanisms and suggested a separation of the Jacobian matrix into two matrices: one associated with the direct kinematics and the other with the inverse kinematics. Depending on which matrix is singular, a closed-loop mechanism may be at a direct kinematic singular configuration, an inverse kinematic singular configuration, or both. This classification has been refined by Zlatanov et al. (1994).

In this chapter, two methods of analysis are presented. The first method employs velocity vector-loop equations, and the second method applies the theory of reciprocal screws.

5.2 JACOBIAN MATRICES

A parallel manipulator such as the VARIAX[®] machining center shown in Fig. 5.1 typically consists of a moving platform and a fixed base connected by several limbs. The moving platform serves as the end effector. Because of the closed-loop construction, not all the joints can be controlled independently. Thus some of the joints are driven by actuators, whereas others are passive.



FIGURE 5.1. VARIAX[®] machining center. (Courtesy of Giddings & Lewis Machine Tools, Fond du Lac, Wisconsin.)

In general, the number of actuated joints should be equal to the number of degrees of freedom of the manipulator.

Let the actuated joint variables be denoted by a vector $\mathbf{q}$ and the location of the moving platform be described by a vector $\mathbf{x}$. Then the kinematic constraints imposed by the limbs can be written in the general form

$$\mathbf{f}(\mathbf{x}, \mathbf{q}) = \mathbf{0}, \quad (5.1)$$

where $\mathbf{f}$ is an n -dimensional implicit function of $\mathbf{q}$ and $\mathbf{x}$ and $\mathbf{0}$ is an n -dimensional zero vector.

Differentiating Eq. (5.1) with respect to time, we obtain a relationship between the input joint rates and the end-effector output velocity as follows:

$$J_x \dot{\mathbf{x}} = J_q \dot{\mathbf{q}}, \quad (5.2)$$

where

$$J_x = \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \quad \text{and} \quad J_q = -\frac{\partial \mathbf{f}}{\partial \mathbf{q}}.$$

The derivation above leads to two separate Jacobian matrices. Hence the overall Jacobian matrix, J , can be written as

$$\dot{\mathbf{q}} = J \dot{\mathbf{x}}, \quad (5.3)$$

where $J = J_q^{-1} J_x$. We note that the Jacobian matrix defined in Eq. (5.3) for a parallel manipulator corresponds to the inverse Jacobian of a serial manipulator.

5.3 SINGULARITY CONDITIONS

Due to the existence of two Jacobian matrices, a parallel manipulator is said to be at a *singular configuration* when either J_x or J_q or both are singular. Three different types of singularities can be identified.

5.3.1 Inverse Kinematic Singularities

An inverse kinematic singularity occurs when the determinant of J_q goes to zero, namely,

$$\det(J_q) = 0. \quad (5.4)$$

When J_q is singular and the null space of J_q is not empty, there exist some nonzero $\dot{\mathbf{q}}$ vectors that result in zero $\dot{\mathbf{x}}$ vectors. Infinitesimal motion of the

moving platform along certain directions cannot be accomplished. Hence the manipulator loses one or more degrees of freedom. On the other hand, at an inverse kinematic singular configuration, a parallel manipulator can resist forces or moments in some directions with zero actuator forces or torques. Inverse kinematic singularities usually occur at the workspace boundary, where different branches of the inverse kinematic solutions converge. It is similar to that of a serial manipulator.

5.3.2 Direct Kinematic Singularities

A direct kinematic singularity occurs when the determinant of J_x is equal to zero, namely,

$$\det(J_x) = 0. \quad (5.5)$$

Assuming that in the presence of such a singular condition the null space of J_x is not empty, there exist some nonzero $\dot{\mathbf{x}}$ vectors that result in zero $\dot{\mathbf{q}}$ vectors. That is, the moving platform can possess infinitesimal motion in some directions while all the actuators are completely locked. Hence the moving platform gains 1 or more degrees of freedom. This is in contradiction with a serial manipulator, which loses 1 or more degrees of freedom (Waldron and Hunt, 1988). In other words, at a direct kinematic singular configuration, the manipulator cannot resist forces or moments in some directions. Direct kinematic singularities usually occur where different branches of direct kinematic solutions meet.

5.3.3 Combined Singularities

A combined singularity occurs when the determinants of J_x and J_q are both zero. Generally, this type of singularity can occur only for manipulators with special kinematic architecture. At a combined singular configuration, Eq. (5.1) will degenerate. The moving platform can undergo some infinitesimal motions while all the actuators are locked. On the other hand, it can also remain stationary while the actuators undergo some infinitesimal motions.

5.4 CONVENTIONAL JACOBIAN

In this section we introduce the conventional method of analysis. Although the methods of screw coordinates and motor algebra are very powerful, their applications to parallel manipulators are often hindered by the presence of many unactuated joints. In this regard, the conventional *velocity vector-loop method* appears to be more straightforward.

Generally, the velocity vector of a point is formulated from two different directions of a loop closure. Each loop closure consists of a fixed base, a moving platform, and all the links of a limb. The unactuated joint rates in each limb are then eliminated by performing a dot product of the velocity vector-loop equation with an appropriate vector that is normal to all vectors of the unactuated joint rates. Finally, the resulting equations are assembled into a Jacobian matrix.

For consistency, we define the end-effector velocity state of the conventional Jacobian as a six-dimensional vector with the linear velocity of a point, followed by the angular velocity of the moving platform; that is,

$$\dot{\mathbf{x}} = \begin{bmatrix} \mathbf{v}_p \\ \boldsymbol{\omega}_p \end{bmatrix}. \quad (5.6)$$

The method can best be illustrated by examples, which we provide next.

5.4.1 Jacobian of a Planar 3RRR Parallel Manipulator

We first investigate the Jacobian and singular conditions of the planar 3RRR manipulator described in Chapter 3. For convenience, a schematic diagram of the manipulator is shown here in Fig. 5.2. The moving platform is defined by the moving pivots A , B , and C , and the fixed platform is defined by the fixed pivots P , Q , and R . Point G is a point in the moving platform, $\mathbf{a}_i$ and $\mathbf{b}_i$

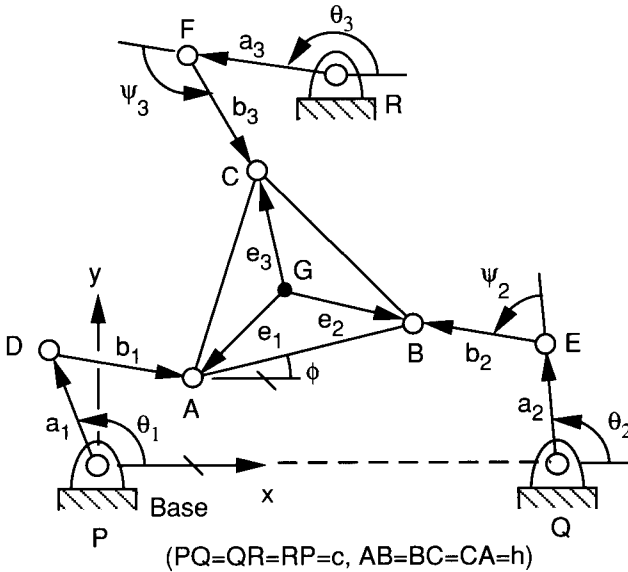


FIGURE 5.2. Schematic of a planar 3RRR manipulator.

denote the vectors of the first and second moving links of the i th limb, while θ_i and ψ_i denote the orientation angles of $\mathbf{a}_i$ and $\mathbf{b}_i$ with respect to the x -axis. For this manipulator, the input vector is $\mathbf{q} = [\theta_1, \theta_2, \theta_3]^T$ and the output vector is $\mathbf{x} = [x_g, y_g, \phi]^T$.

A loop-closure equation can be written for each limb. For example, for the first limb we have

$$\overline{PG} + \overline{GA} = \overline{PD} + \overline{DA}. \quad (5.7)$$

Since this is a planar mechanism, the angular velocity vectors of all links point in the positive z -direction. A velocity vector-loop equation is obtained by taking the derivative of Eq. (5.7) with respect to time:

$$\mathbf{v}_g + \dot{\phi}(\mathbf{k} \times \mathbf{e}_i) = \dot{\theta}_i(\mathbf{k} \times \mathbf{a}_i) + (\dot{\theta}_i + \dot{\psi}_i)(\mathbf{k} \times \mathbf{b}_i), \quad i = 1, 2, 3, \quad (5.8)$$

where $\mathbf{v}_g$ is the velocity of point G and $\mathbf{k}$ is a unit vector pointing in the positive z -axis direction. Since $\dot{\psi}_i$ is a passive variable, it should be eliminated from Eq. (5.8). To achieve this goal, we dot-multiply both sides of Eq. (5.8) by $\mathbf{b}_i$. This leads to

$$\mathbf{b}_i \cdot \mathbf{v}_g + \dot{\phi} \mathbf{k} \cdot (\mathbf{e} \times \mathbf{b}_i) = \dot{\theta}_i \mathbf{k} \cdot (\mathbf{a}_i \times \mathbf{b}_i). \quad (5.9)$$

Writing Eq. (5.9) three times, once for each $i = 1, 2$, and 3 , yields three scalar equations, which can be arranged in matrix form:

$$J_x \dot{\mathbf{x}} = J_q \dot{\mathbf{q}}, \quad (5.10)$$

where

$$J_x = \begin{bmatrix} b_{1x} & b_{1y} & e_{1x}b_{1y} - e_{1y}b_{1x} \\ b_{2x} & b_{2y} & e_{2x}b_{2y} - e_{2y}b_{2x} \\ b_{3x} & b_{3y} & e_{3x}b_{3y} - e_{3y}b_{3x} \end{bmatrix},$$

$$J_q = \begin{bmatrix} a_{1x}b_{1y} - a_{1y}b_{1x} & 0 & 0 \\ 0 & a_{2x}b_{2y} - a_{2y}b_{2x} & 0 \\ 0 & 0 & a_{3x}b_{3y} - a_{3y}b_{3x} \end{bmatrix},$$

and where $\dot{\mathbf{x}} = [v_{gx}, v_{gy}, \dot{\phi}]^T$ and $\dot{\mathbf{q}} = [\dot{\theta}_1, \dot{\theta}_2, \dot{\theta}_3]^T$. In what follows, we discuss the physical significance of each of the three types of singularities.

(a) Inverse Kinematic Singularities. Inverse kinematic singularities occur when one of the diagonal elements of J_q vanishes, namely,

$$a_{ix}b_{iy} - a_{iy}b_{ix} = 0 \quad \text{for } i = 1, \text{ or } 2, \text{ or } 3. \quad (5.11)$$

The left-hand side of Eq. (5.11) represents the magnitude of $\mathbf{a}_i \times \mathbf{b}_i$. Consequently, an inverse kinematic singularity arises whenever any limb is in a fully stretched-out or folded-back configuration. The manipulator loses 1, 2, or 3 degrees of freedom, depending on whether one, two, or three limbs are in fully stretched-out or folded-back configurations. At an inverse kinematic singularity, infinitesimal rotation of the input link, which is in a stretched-out or folded-back configuration, results in no output motion of the moving platform.

(b) Direct Kinematic Singularities. Direct kinematic singularities occur when the determinant of J_x goes to zero. Although it is difficult to identify all possible direct kinematic singularities, the following two singularities are found by inspection of the matrix J_x . We note that the last column of J_x represents the magnitudes of the vector cross products, $\mathbf{e}_i \times \mathbf{b}_i$ for $i = 1, 2$, and 3. These three elements will vanish when the vector $\mathbf{b}_i$ is in line with $\mathbf{e}_i$ for all the limbs. Since G is an arbitrary point in the moving platform, a direct kinematic singularity arises whenever the three vectors $\mathbf{b}_i$, for $i = 1, 2$, and 3, intersect at a common point. Figure 5.3 illustrates such a singular configuration. At this configuration, the moving platform can make an infinitesimal rotation about point G while the actuators are locked. Hence the moving platform gains 1 degree of freedom and it cannot withstand any external moment about G .

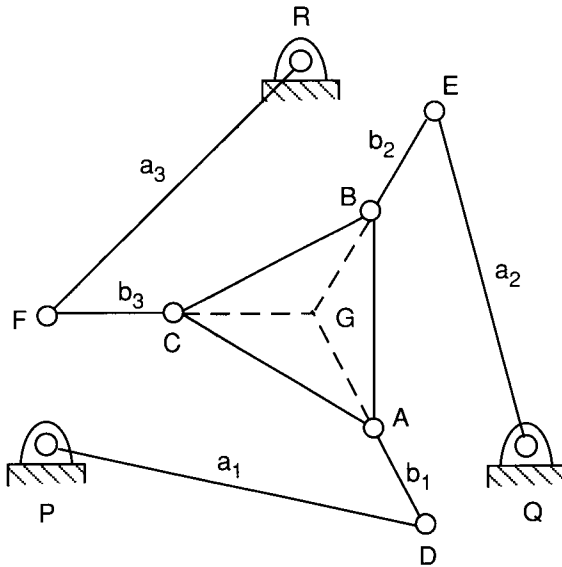


FIGURE 5.3. Direct kinematic singularity of the planar parallel manipulator.

We also note that the first and second columns of J_x represent the x and y components of the vector $\mathbf{b}_i$, for $i = 1, 2$, and 3 . These two columns become linearly dependent when the three vectors, $\mathbf{b}_i$ for $i = 1, 2$, and 3 , point in the same direction. Hence another direct kinematic singularity occurs whenever the three vectors $\mathbf{b}_i$, for $i = 1, 2$, and 3 , are parallel to one another. Figure 5.4 illustrates such a singular configuration. At this configuration, the moving platform can make an infinitesimal translation along a direction that is perpendicular to the vector $\mathbf{b}_i$ while all the actuators are locked. Any force applied to the moving platform in that direction cannot be resisted by the actuators.

(c) Combined Singularities. A combined singularity occurs when the determinants of J_x and J_q are both equal to zero. As mentioned earlier, this type of singularity is not only configuration dependent but also architecture dependent. Each of the following two architectures possesses a combined singularity (Gosselin and Angeles, 1990).

Architecture 1

$$PQ = QR = RP, \quad AB = BA = CA, \quad a_i = PQ/\sqrt{3},$$

$$\text{and } b_i = e_i \quad \text{for } i = 1, 2, \text{ and } 3.$$

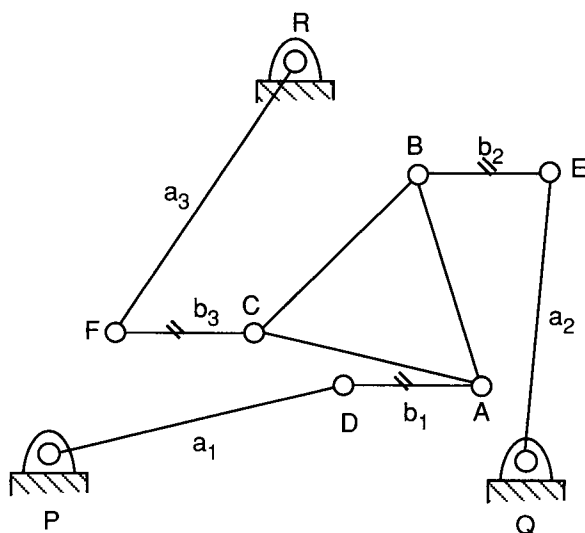


FIGURE 5.4. Another direct kinematic singularity of the planar parallel manipulator.

This type of manipulator can be positioned in a configuration in which pivots D , E , and F meet at the centroid of the base triangle. Because of the special link length ratios, the centroid of the moving platform also coincides with the centroid of the base triangle. Therefore, the moving platform can rotate about the centroid while the actuators are locked. On the other hand, the moving platform can be held stationary at $\phi = 0$ while the input links make some infinitesimal rotations about their corresponding pivots. This is due to the fact that the elements of J_q and the last column of J_x are all equal to zero.

Architecture 2

$$PQ = QR = RP = AB = BC = CA$$

$$\text{and } a_i = b_i \quad \text{for } i = 1, 2, \text{ and } 3.$$

A manipulator of this type can be positioned in a configuration in which the three moving pivots A , B , and C are coincident with the three fixed pivots P , Q , and R , respectively. Hence the three input links can make arbitrary rotations while the moving platform is locked. Furthermore, with the actuators locked at $\theta_1 = -150^\circ$, $\theta_2 = -30^\circ$, and $\theta_3 = 90^\circ$, the moving platform can perform an infinitesimal rotation about its centroid.

5.4.2 Jacobian of a Spatial Orientation Mechanism

In this example we study the Jacobian and singular conditions of the spatial orientation mechanism shown in Fig. 3.7. Since the manipulator possesses only 3 rotational degrees of freedom, the input vector can be written as $\dot{\mathbf{q}} = [\dot{d}_1, \dot{d}_2, \dot{d}_3]^T$, and the output vector can be described by the angular velocity of the moving platform, $\dot{\mathbf{x}} = \boldsymbol{\omega}_B = [\omega_x, \omega_y, \omega_z]^T$.

Referring to Fig. 3.7, a loop-closure equation for the i th limb can be written as

$$\overline{OB_i} = \overline{OA_i} + \overline{A_iB_i}. \quad (5.12)$$

Taking the derivative of Eq. (5.12) with respect to time yields a velocity vector-loop equation as follows:

$$\boldsymbol{\omega}_B \times \mathbf{b}_i = d_i \boldsymbol{\omega}_i \times \mathbf{s}_i + \dot{d}_i \mathbf{s}_i \quad \text{for } i = 1, 2, 3, \quad (5.13)$$

where $\mathbf{b}_i$ denotes the vector of $\overline{OB_i}$, $\mathbf{s}_i$ is a unit vector pointing along $\overline{A_iB_i}$, and $\boldsymbol{\omega}_i$ denotes the angular velocity of limb i with respect to the fixed frame.

To eliminate ω_i , we dot-multiply both sides of Eq. (5.13) by $\mathbf{s}_i$. This produces

$$(\mathbf{b}_i \times \mathbf{s}_i) \cdot \omega_B = \dot{d}_i \quad \text{for } i = 1, 2, 3. \quad (5.14)$$

Equation (5.14) written three times, once for each $i = 1, 2$, and 3 , yields three scalar equations, which can be arranged in matrix form:

$$J_x \omega_B = J_q \dot{\mathbf{q}}, \quad (5.15)$$

where

$$J_x = \begin{bmatrix} (\mathbf{b}_1 \times \mathbf{s}_1)^T \\ (\mathbf{b}_2 \times \mathbf{s}_2)^T \\ (\mathbf{b}_3 \times \mathbf{s}_3)^T \end{bmatrix},$$

$$J_q = I \quad (3 \times 3 \text{ identity matrix}).$$

Note that each row of J_x represents a vector that is normal to a plane defined by the triangle $\triangle A_i O B_i$. For convenience, we define this vector as $\mathbf{n}_i$; that is,

$$\mathbf{n}_i = \mathbf{b}_i \times \mathbf{s}_i. \quad (5.16)$$

In what follows, we discuss the physical meaning of the three types of singularities.

(a) Inverse Kinematic Singularities. Since J_q is an identity matrix, there exists no inverse kinematic singularity within the workspace of the manipulator. However, inverse kinematic singularities can occur at the workspace boundary where one or more limbs are in fully stretched or retracted positions.

(b) Direct Kinematic Singularities. Consider the kinematic constraints imposed by limb i . From Eq. (5.14), we observe that the elongation rate of the i th limb, $\dot{d}_i$, contributes to a rotation of the moving platform about an axis that passes through the origin O and points in the direction of the vector $\mathbf{n}_i$. Thus when the i th actuator is locked, the moving platform is prohibited from rotating about this axis. However, it can still rotate freely about the two axes defined by $\overline{OA_i}$ and $\overline{OB_i}$. To completely immobilize the moving platform by three linear actuators, the three vectors $\mathbf{n}_i$ for $i = 1, 2$, and 3 must be well defined and span the three-dimensional rotation space of the end effector. Hence the determinant of J_x goes to zero and the manipulator gains an extra degree of freedom, when these three vectors become linearly dependent or when one of the three vectors vanishes identically. Three different direct kinematic singularities can be identified.

Case 1. One of the three vectors, say $\mathbf{n}_i$, vanishes. For this condition to hold, points A_i , O , and B_i must lie on a straight line. Under such a condition, the manipulator gains 1 degree of freedom. That is, with all actuators locked, the moving platform can make an infinitesimal rotation about a line of intersection of the two planes defined by triangles $\triangle A_j O B_j$ and $\triangle A_k O B_k$, where $i \neq j \neq k$. Figure 5.5 shows such a singular configuration, in which points A_1 , O , and B_1 lie on a straight line.

Case 2. Two of the three vectors, say $\mathbf{n}_i$ and $\mathbf{n}_j$, are linearly dependent. When the two planes defined by triangles $\triangle A_i O B_i$ and $\triangle A_j O B_j$ are coincident, the manipulator gains 1 degree of freedom. Specifically, with all actuators locked, the moving platform can make an infinitesimal rotation about a line of intersection of the plane defined by the triangle $\triangle A_i O B_i$ and the plane defined by the triangle $\triangle A_k O B_k$, where $k \neq j$. Figure 5.6 shows such a singular configuration, where the two faces $\triangle A_1 O B_1$ and $\triangle A_2 O B_2$ lie on a plane.

Case 3. The three vectors $\mathbf{n}_1$, $\mathbf{n}_2$, and $\mathbf{n}_3$ are linearly dependent. This condition happens when the three planes defined by triangles $\triangle A_1 O B_1$, $\triangle A_2 O B_2$, and $\triangle A_3 O B_3$ intersect in a common line. In other words, when the three

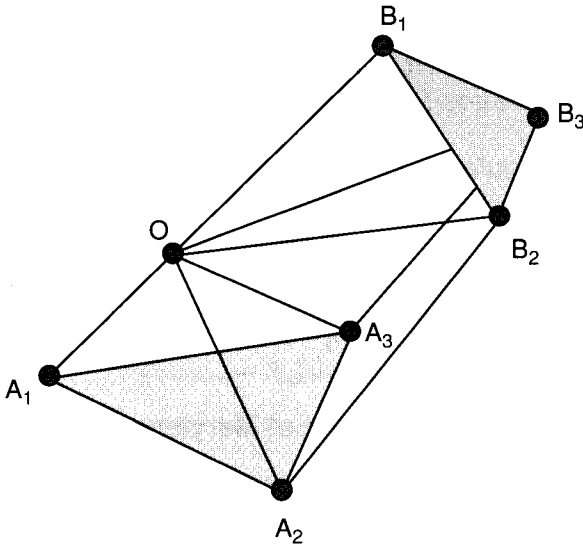


FIGURE 5.5. Direct kinematic singularity of the orientation mechanism.

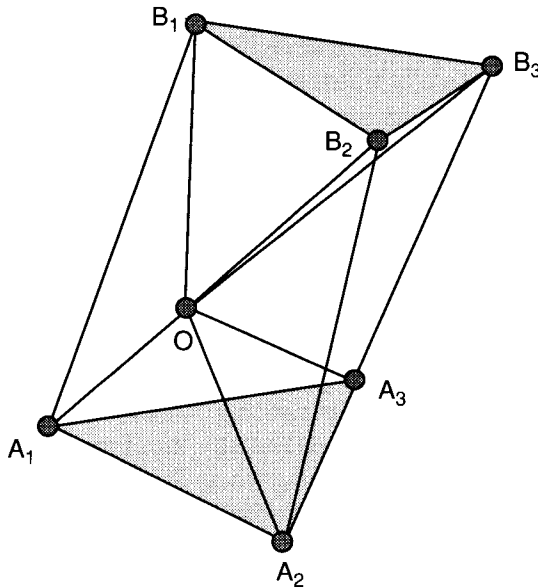


FIGURE 5.6. Another direct kinematic singularity of the orientation mechanism.

normal vectors $\mathbf{n}_1$, $\mathbf{n}_2$, and $\mathbf{n}_3$ lie on a plane, one of the three vectors can be expressed as a linear summation of the other two. Hence, with all actuators locked, the moving platform can make an infinitesimal rotation about the line of intersection of the three planes. For example, if the moving tetrahedron $OB_1B_2B_3$ is geometrically similar to the fixed tetrahedron $OA_1A_2A_3$, a singular condition occurs when the moving tetrahedron $OB_1B_2B_3$ is directly on top of the fixed tetrahedron $OA_1A_2A_3$.

(c) Combined Singularities. Combined singularities cannot occur within the workspace of the manipulator. However, it can occur at the workspace boundary if the following three conditions are satisfied simultaneously: (1) the tetrahedron $OA_1A_2A_3$ is similar to $OB_1B_2B_3$; (2) the three planes of triangle $\triangle A_iOB_i$, for $i = 1, 2, 3$, intersect at a common line; and (3) one of the three limbs is at its extreme reach.

5.4.3 Jacobian of the University of Maryland's Manipulator

In this example we develop the Jacobian matrix and examine some singular configurations of the University of Maryland manipulator shown in Fig. 3.10. Recall that this manipulator possesses only 3 translational degrees of freedom. Referring to Fig. 3.11, a loop-closure equation for the i th limb can be

written as

$$\overline{OP} + \overline{PC_i} = \overline{OA_i} + \overline{A_iB_i} + \overline{B_iC_i}. \quad (5.17)$$

Differentiating Eq. (5.17) with respects to time yields

$$\mathbf{v}_p = \boldsymbol{\omega}_{1i} \times \mathbf{a}_i + \boldsymbol{\omega}_{2i} \times \mathbf{b}_i, \quad (5.18)$$

where $\mathbf{v}_p$ is the linear velocity of the moving platform, $\mathbf{a}_i = \overline{A_iB_i}$, $\mathbf{b}_i = \overline{B_iC_i}$, and $\boldsymbol{\omega}_{ji}$ is the angular velocity of the j th link of the i th limb. Here link A_iB_i is treated as the first link and B_iC_i as the second link.

For this manipulator, the input vector is $\dot{\mathbf{q}} = [\dot{\theta}_{11}, \dot{\theta}_{12}, \dot{\theta}_{13}]^T$, and the output vector is $\mathbf{v}_p = [v_{p,x}, v_{p,y}, v_{p,z}]^T$. All other joint rates are passive variables. To eliminate the passive joint rate, we dot-multiply both sides of Eq. (5.18) by $\mathbf{b}_i$. This produces

$$\mathbf{b}_i \cdot \mathbf{v}_p = \boldsymbol{\omega}_{1i} \cdot (\mathbf{a}_i \times \mathbf{b}_i). \quad (5.19)$$

Expressing the vectors in Eq. (5.19) in the (x_i, y_i, z_i) coordinate frame, we have

$${}^i\mathbf{a}_i = a \begin{bmatrix} c\theta_{1i} \\ 0 \\ s\theta_{1i} \end{bmatrix}, \quad {}^i\mathbf{b}_i = b \begin{bmatrix} s\theta_{3i}c(\theta_{1i} + \theta_{2i}) \\ c\theta_{3i} \\ s\theta_{3i}s(\theta_{1i} + \theta_{2i}) \end{bmatrix}, \quad {}^i\boldsymbol{\omega}_{1i} = \begin{bmatrix} 0 \\ -\dot{\theta}_{1i} \\ 0 \end{bmatrix},$$

$${}^i\mathbf{v}_p = \begin{bmatrix} v_{p,x}c\phi_i + v_{p,y}s\phi_i \\ -v_{p,x}s\phi_i + v_{p,y}c\phi_i \\ v_{p,z} \end{bmatrix}.$$

Substituting the expressions above into Eq. (5.19), after simplification we obtain

$$j_{ix}v_{p,x} + j_{iy}v_{p,y} + j_{iz}v_{p,z} = as\theta_{2i}s\theta_{3i}\dot{\theta}_{1i}, \quad (5.20)$$

where

$$j_{ix} = c(\theta_{1i} + \theta_{2i})s\theta_{3i}c\phi_i - c\theta_{3i}s\phi_i,$$

$$j_{iy} = c(\theta_{1i} + \theta_{2i})s\theta_{3i}s\phi_i + c\theta_{3i}c\phi_i,$$

$$j_{iz} = s(\theta_{1i} + \theta_{2i})s\theta_{3i}.$$

Note that $\mathbf{j}_i = [j_{ix}, j_{iy}, j_{iz}]^T$ represents a unit vector that is directed from B_i to C_i and expressed in the fixed (x, y, z) coordinate frame.

Writing Eq. (5.20) three times, once for each $i = 1, 2$, and 3 , yields three scalar equations, which can be assembled in matrix form as

$$J_x \mathbf{v}_p = J_q \dot{\mathbf{q}}, \quad (5.21)$$

where

$$J_x = \begin{bmatrix} \dot{j}_{1x} & \dot{j}_{1y} & \dot{j}_{1z} \\ \dot{j}_{2x} & \dot{j}_{2y} & \dot{j}_{2z} \\ \dot{j}_{3x} & \dot{j}_{3y} & \dot{j}_{3z} \end{bmatrix},$$

$$J_q = a \begin{bmatrix} s\theta_{21}s\theta_{31} & 0 & 0 \\ 0 & s\theta_{22}s\theta_{32} & 0 \\ 0 & 0 & s\theta_{23}s\theta_{33} \end{bmatrix}.$$

(a) Inverse Kinematic Singularities. From Eq. (5.21), we observe that an inverse kinematic singularity occurs when one of the following conditions is satisfied:

$$\theta_{2i} = 0 \quad \text{or} \quad \pi, \quad (5.22)$$

or

$$\theta_{3i} = 0 \quad \text{or} \quad \pi, \quad (5.23)$$

for $i = 1$ or 2 or 3 . Physically, the condition $\theta_{2i} = 0$ or π occurs when the upper arm linkage and the input link of a limb are in the same plane. The condition $\theta_{3i} = 0$ or π occurs when all the links of the four-bar parallelogram that comprise the upper arm of a limb are collinear.

(b) Direct Kinematic Singularities. Direct kinematic singularities occur when the condition $\det(J_x) = 0$ is satisfied. Although it is difficult to solve for all possible singularities, several manipulator postures that satisfy this condition are identified below. We note that each row of J_x represents a unit vector, $\mathbf{j}_i$, which points along the direction of a coupling link B_iC_i of the four-bar parallelogram. Hence J_x becomes singular when these three unit vectors become linearly dependent; that is,

$$\mu_1 \mathbf{j}_1 + \mu_2 \mathbf{j}_2 + \mu_3 \mathbf{j}_3 = \mathbf{0} \quad (5.24)$$

for some real values of μ_1 , μ_2 , and μ_3 , where not all μ 's are zero.

One condition that satisfies Eq. (5.24) is found when all three unit vectors $\mathbf{j}_i$ for $i = 1$ to 3 lie on a plane. For example, when the three four-bar parallelograms lie on a plane that is parallel to the x - y plane, the z -component of Eq. (5.24) is equal to zero identically. Hence some nonzero values of μ_1 , μ_2 , and μ_3 can always be found from the first two equations. In terms of manip-

ulator variables, this condition can be written as

$$s(\theta_{11} + \theta_{21})s\theta_{31} = s(\theta_{12} + \theta_{22})s\theta_{32} = s(\theta_{13} + \theta_{23})s\theta_{33} = 0. \quad (5.25)$$

Equation (5.25) indicates that the manipulator is in a direct kinematic singularity whenever all limbs are in a posture such that

$$\theta_{1i} + \theta_{2i} = 0 \quad \text{or} \quad \pi,$$

or

$$\theta_{3i} = 0 \quad \text{or} \quad \pi,$$

for all $i = 1$ and 2 and 3 . The condition $\theta_{3i} = 0$ or π is a combined singularity.

The geometrical interpretation of this condition is that the manipulator displays a direct kinematic singularity when the four-bar linkages of all three limbs are in the same plane as the moving platform. At such a configuration, the manipulator actuators cannot resist any force applied to the moving platform in the z -direction. One example of this type of singular configuration is shown in Fig. 5.7. Note that this example configuration is possible only if $a + r \geq b + h$.

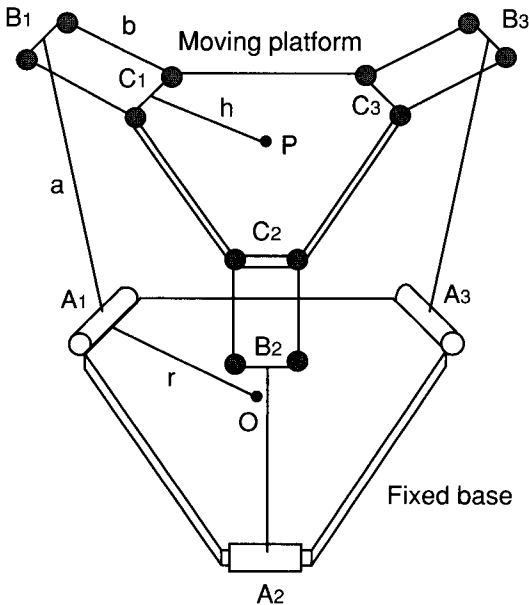


FIGURE 5.7. Direct kinematic singular configuration in which all the upper arm linkages are in the plane of the moving platform.

A second set of postures that satisfies Eq. (5.24) is found when any two of the upper arm linkages are parallel; that is,

$$\mathbf{j}_i = \pm \mathbf{j}_k \quad \text{for } i \neq k. \quad (5.26)$$

Figure 5.8 shows such a singular configuration. At this singular configuration, the manipulator cannot resist any force applied in the plane of the moving platform. Note that in this example we have sketched all three upper arm linkages to be parallel to one another. However, it is only necessary that two of the upper arm linkages be parallel to each other to yield a singular configuration.

(c) Combined Singularities. One combined singularity occurs when the following two conditions are satisfied simultaneously: (1) the geometry of the moving platform is identical to that of the fixed platform, and (2) all three input links are perpendicular to the fixed base (i.e., $\overline{A_i B_i} \perp \overline{O A_i}$ for $i = 1, 2$, and 3). At this configuration the moving platform gains 2 degrees of freedom. Specifically, with the input links completely locked, point P can be positioned anywhere on a spherical surface centered at a distance a above the centroid of

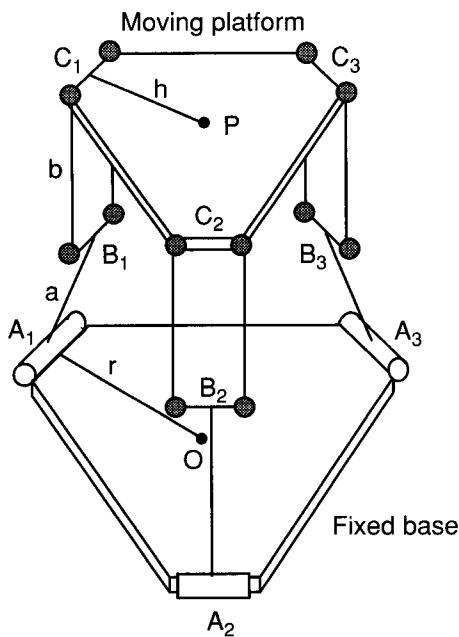


FIGURE 5.8. Another direct kinematic singular configuration, in which two upper arm linkages are parallel to each other.

the fixed base. On the other hand, with the moving platform held stationary, the input links can make some infinitesimal rotations.

5.4.4 Jacobian of a Stewart–Gough Platform

In this example we investigate the Jacobian and singular conditions of the Stewart–Gough platform shown in Fig. 3.15. For this manipulator, the input vector is given by $\dot{\mathbf{q}} = [\dot{d}_1, \dot{d}_2, \dots, \dot{d}_6]^T$, and the output vector can be described by the velocity of the centroid P and the angular velocity of the moving platform:

$$\dot{\mathbf{x}} = \begin{bmatrix} \mathbf{v}_p \\ \boldsymbol{\omega}_B \end{bmatrix}. \quad (5.27)$$

The Jacobian matrix can be derived by formulating a velocity loop-closure equation for each limb. Referring to Fig. 3.15, a loop-closure equation for the i th limb can be written as

$$\overline{OP} + \overline{PB_i} = \overline{OA_i} + \overline{A_iB_i}. \quad (5.28)$$

Differentiating Eq. (5.28) with respect to time yields

$$\mathbf{v}_p + \boldsymbol{\omega}_B \times \mathbf{b}_i = d_i \boldsymbol{\omega}_i \times \mathbf{s}_i + \dot{d}_i \mathbf{s}_i, \quad (5.29)$$

where $\mathbf{b}_i$ and $\mathbf{s}_i$ denote the vector $\overline{PB_i}$ and a unit vector along $\overline{A_iB_i}$, respectively, and $\boldsymbol{\omega}_i$ denotes the angular velocity of the i th limb with respect to the fixed frame A . To eliminate $\boldsymbol{\omega}_i$, we dot-multiply both sides of Eq. (5.29) by $\mathbf{s}_i$:

$$\mathbf{s}_i \cdot \mathbf{v}_p + (\mathbf{b}_i \times \mathbf{s}_i) \cdot \boldsymbol{\omega}_B = \dot{d}_i. \quad (5.30)$$

Equation (5.30) written six times, once for each $i = 1$ to 6, yields six scalar equations, which can be assembled in matrix form:

$$J_x \dot{\mathbf{x}} = J_q \dot{\mathbf{q}}, \quad (5.31)$$

where

$$J_x = \begin{bmatrix} \mathbf{s}_1^T & (\mathbf{b}_1 \times \mathbf{s}_1)^T \\ \mathbf{s}_2^T & (\mathbf{b}_2 \times \mathbf{s}_2)^T \\ \vdots & \vdots \\ \mathbf{s}_6^T & (\mathbf{b}_6 \times \mathbf{s}_6)^T \end{bmatrix},$$

$$J_q = I \quad (6 \times 6 \text{ identity matrix})$$

Let

$$\mathbf{n}_i = \mathbf{b}_i \times \mathbf{s}_i. \quad (5.32)$$

Then $\mathbf{n}_i$ represents a vector that is normal to the plane containing points A_i , B_i , and P . In what follows, we discuss the physical meaning of the three types of singularities.

(a) Inverse Kinematic Singularities. Inverse kinematic singularities cannot occur within the workspace of the manipulator since J_q is an identity matrix. However, inverse kinematic singularities can occur at the workspace boundary where one or more limbs are in fully stretched or retracted positions.

(b) Direct Kinematic Singularities. This type of singularity is very difficult to solve (Merlet, 1989; St-Onge and Gosselin, 1996). Although it is impossible to find all singularities, some of them can be identified by an examination of the matrix J_x . For example, when the moving platform falls on top of the fixed base, $\mathbf{n}_i = 0$, identically, for $i = 1$ to 6. The manipulator gains 3 degrees of freedom. That is, the moving platform can make an infinitesimal rotation about any axis on the x - y plane. It can also make an infinitesimal translation along the z -axis.

Another direct kinematic singularity occurs when the geometry of the manipulator satisfies the following conditions:

1. The geometry of the moving platform is identical to that of the fixed platform.
2. The manipulator assumes a configuration in which the limb lengths are equal to one another.
3. All the limbs are parallel to each other.

When the foregoing conditions are met, the moving platform gains 2 translational degrees of freedom. Namely, it can be positioned anywhere on a spherical surface of a radius equal to the length of a limb and centered at the geometric center of the fixed platform. The orientation of the moving platform remains unchanged throughout this motion. There are infinitely many such configurations, whenever the condition $d_1 = d_2 = \dots = d_6$ is met. Finally, we note that a combined singularity cannot occur within the workspace of the manipulator since J_q is an identity matrix.

5.5 WRENCHES AND RECIPROCAL SCREWS

The concept of twist is a convenient way of describing either finite or infinitesimal displacements of a rigid body in three-dimensional space. For statics, a similar concept can be defined. It is well known that any system of forces and couples acting on a rigid body can be reduced to a resultant force and a couple about any point of interest. In general, the resultant force vector and the couple are not collinear. However, it can be shown that there exists a unique axis with respect to which the system of forces and couples can be reduced to a resulting force, $\mathbf{f}$, acting along the axis and a couple, $\mathbf{c}$, about the same axis. This force and couple combination is called a *wrench* (Roth, 1984; Yuan et al., 1971). The unique axis is called the *wrench axis* or *screw axis* of the system of forces and couples. Similar to a twist, the *pitch* of a wrench is defined as the ratio of the couple to the force (i.e., $\lambda_r = c/f$).

A unit wrench, $\hat{\$}_r$, is defined with respect to a reference frame by a pair of vectors:

$$\hat{\$}_r = \begin{bmatrix} \mathbf{s}_r \\ \mathbf{s}_{ro} \times \mathbf{s}_r + \lambda_r \mathbf{s}_r \end{bmatrix} = \begin{bmatrix} S_{r1} \\ S_{r2} \\ S_{r3} \\ S_{r4} \\ S_{r5} \\ S_{r6} \end{bmatrix}, \quad (5.33)$$

where $\mathbf{s}_r$ is a unit vector pointing along the direction of the screw axis, $\mathbf{s}_{ro}$ is the position vector of any point on the screw axis, and the vector $\mathbf{s}_{ro} \times \mathbf{s}_r$ defines the moment of the screw axis about the origin of the reference frame chosen. For a pure force, $\lambda_r = 0$, the unit wrench reduces to

$$\hat{\$}_r = \begin{bmatrix} \mathbf{s}_r \\ \mathbf{s}_{ro} \times \mathbf{s}_r \end{bmatrix}. \quad (5.34)$$

For a pure couple, $\lambda_r = \infty$, the unit wrench is defined as

$$\hat{\$}_r = \begin{bmatrix} \mathbf{0} \\ \mathbf{s}_r \end{bmatrix}. \quad (5.35)$$

Following the definition above, a wrench of intensity ρ can be written in terms of the unit wrench as

$$\$_r = \rho \hat{\$}_r. \quad (5.36)$$

We observe that the first three components of a wrench represent the resulting force, $\mathbf{f}$, and the last three components represent the resulting moment due to the combined effects of the force $\mathbf{f}$ and the couple $\mathbf{c}$ about the origin of the reference frame.

The concept of reciprocal screws was first studied by Ball (1900), followed by Waldron (1969), Hunt (1970, 1978), Roth (1984), and others. If a wrench acts on a rigid body in such a way that it produces no work while the body is undergoing an infinitesimal twist, the two screws are said to be *reciprocal screws*. Figure 5.9 shows a rigid body that is exerted by a wrench $\$r = \rho\hat{\$}_r$ while it is undergoing an infinitesimal twist $\$ = \dot{q}\hat{\$}$. Recall that the first three components of a twist represent the angular velocity and the last three components represent the linear velocity of a point that is instantaneously coincident with the origin of the reference frame. Hence the virtual work performed between the wrench and the twist is given by

$$\begin{aligned}\delta W &= \rho \dot{q} [\mathbf{s} \cdot (\mathbf{s}_{ro} \times \mathbf{s}_r + \lambda_r \mathbf{s}_r) + \mathbf{s}_r \cdot (\mathbf{s}_o \times \mathbf{s} + \lambda \mathbf{s})] \\ &= \rho \dot{q} [(\lambda + \lambda_r)(\mathbf{s} \cdot \mathbf{s}_r) + \mathbf{s}_r \cdot (\mathbf{s}_o \times \mathbf{s}) + \mathbf{s} \cdot (\mathbf{s}_{ro} \times \mathbf{s}_r)].\end{aligned}\quad (5.37)$$

From the geometry of the lines associated with the two screws, it can be shown that

$$\mathbf{s} \cdot \mathbf{s}_r = c\alpha, \quad (5.38)$$

$$\mathbf{s}_r \cdot (\mathbf{s}_o \times \mathbf{s}) + \mathbf{s} \cdot (\mathbf{s}_{ro} \times \mathbf{s}_r) = -\mathbf{a} \cdot (\mathbf{s} \times \mathbf{s}_r) = -a s \alpha, \quad (5.39)$$

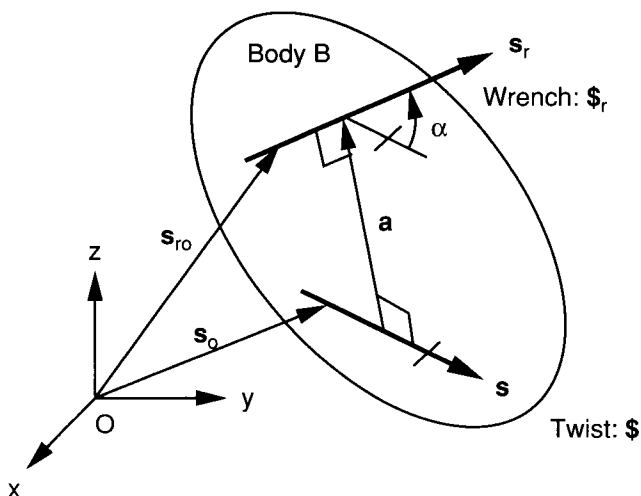


FIGURE 5.9. Twist and wrench in three-dimensional space.

where $\mathbf{a}$ is a vector along the common perpendicular leading from the screw axis of $\$$ to $\$_r$ and α is the twist angle between the axes of $\$$ and $\$_r$, measured from $\$$ to $\$_r$ about the common perpendicular, $\mathbf{a}$, according to the right-hand rule. Substituting Eqs. (5.38) and (5.39) into (5.37), we obtain

$$\delta W = \rho \dot{q}[(\lambda + \lambda_r)c\alpha - a s \alpha]. \quad (5.40)$$

By definition, the virtual work produced by the two reciprocal screws is equal to zero. Since ρ and $\dot{q}$ are generally not equal to zero, the reciprocal condition can be stated as

$$\eta \equiv (\lambda + \lambda_r)c\alpha - a s \alpha = 0. \quad (5.41)$$

The quantity η in Eq. (5.41) is called the *virtual coefficient* between the twist and wrench. We note that the virtual coefficient is independent of the intensities of the twist and wrench. If $\eta = 0$, then regardless of the intensities of the twist and wrench, the wrench will not produce any work while the rigid body undergoes an infinitesimal twist. Since the virtual coefficient is symmetrical in λ and λ_r , the two screws $\$$ and $\$_r$ can be interchanged without affecting their reciprocal properties.

For convenience, we define the *transpose of a screw* as

$$\$^T = [S_4 \ S_5 \ S_6 \ S_1 \ S_2 \ S_3],$$

such that the *generalized inner product* or the *orthogonal product* of two screws $\$_r$ and $\$$ is given by

$$\$_r^T \$ = S_1 S_{r4} + S_2 S_{r5} + S_3 S_{r6} + S_4 S_{r1} + S_5 S_{r2} + S_6 S_{r3}.$$

Then the reciprocal condition can be stated as

$$\delta W = \$_r^T \$ = 0. \quad (5.42)$$

Equation (5.41) or (5.42) imposes one geometric constraint on the two reciprocal screws. Since a unit screw requires five independent parameters to specify its location and pitch, there is a quadruple infinitude (∞^4) of screws reciprocal to a given screw. Thus all screws that are reciprocal to a single screw form a 5-system in three-dimensional space. Theoretically, any five screws selected from the 5-system can be used as “wrenches” to constrain the rigid body to a single-degree-of-freedom motion. Similarly, there is a triple infinitude (∞^3) of screws reciprocal to the single infinitude of screws associated with a 2-system, and these reciprocal screws form a 4-system. Any four screws selected from the 4-system can be used as “wrenches” to constrain

the rigid body to a 2-dof motion. The 3-system and its reciprocal system both contain a double infinitude (∞^2) of screws. A more detailed description of screw systems can be found in Hunt (1970).

In general, the condition for reciprocity is satisfied if the point of application of a wrench remains stationary while the rigid body is undergoing an instantaneous twist about some axis. For a wrench of zero pitch (i.e., a pure force) the condition is satisfied if either the contact point does not move at all or moves in a direction orthogonal to the applied force.

In what follows, we study the reciprocal screw systems associated with some frequently used kinematic pairs and chains that can be used as guidelines for the development of Jacobian matrices of parallel manipulators.

5.5.1 Reciprocal Screws of Some Kinematic Pairs

The screws and reciprocal screws associated with some frequently used joints are listed below.

Revolute Joint. The unit screw associated with a revolute joint is a screw of zero pitch pointing along the joint axis. The reciprocal screws form a 5-system. In particular, those zero-pitch reciprocal screws lie on all planes containing the axis of the revolute joint.

Prismatic Joint. The unit screw associated with a prismatic joint is a screw of infinite pitch pointing along the direction of sliding. The reciprocal screws form a 5-system, and those zero-pitch reciprocal screws lie on all planes perpendicular to the axis of the prismatic joint.

Spherical Joint. The unit screws associated with a spherical joint form a 3-system of zero pitch passing through the center of the joint. The reciprocal screws also form a 3-system of zero pitch passing through the center of the sphere.

Universal Joint. The unit screws associated with a universal joint form a 2-system of zero pitch. It is a planar pencil radiating from the center of the universal joint and lying on a plane that contains the two axes of revolution. The reciprocal screws form a 4-system. All zero-pitch reciprocal screws either pass through the center of the universal joint or lie on a plane defined by the axes of the universal joint. Furthermore, there exists an infinite-pitch reciprocal screw that passes through the center of the universal joint and is perpendicular to both joint axes.

5.5.2 Reciprocal Screws of Some Kinematic Chains

Reciprocal screws associated with a kinematic chain can be found by an intersection of the systems of reciprocal screws associated with the joints of the kinematic chain.

Universal–Spherical Dyad. The joint screws associated with a universal–spherical dyad form a 5-system. Hence the reciprocal screw is a 1-system. Because of the presence of a spherical joint and a universal joint, the reciprocal screw is a zero-pitch screw passing through the centers of the two joints.

Revolute–Spherical Dyad. The joint screws associated with a revolute–spherical dyad form a 4-system. The reciprocal screws form a 2-system. Because of the presence of a spherical joint, all reciprocal screws are zero-pitch screws passing through the center of the spherical joint. Due to the presence of a revolute joint, the reciprocal screws must also lie on a plane containing the axis of the revolute joint. Hence all the reciprocal screws are zero-pitch screws forming a planar pencil, as shown in Fig. 5.10.

Prismatic–Spherical Dyad. The joint screws associated with a prismatic–spherical dyad form a 4-system. The reciprocal screws form a 2-system. Because of the presence of a spherical joint, all reciprocal screws are zero-pitch screws passing through the center of the sphere. Furthermore, due to the presence of a prismatic joint, the reciprocal screws must also lie on a plane that

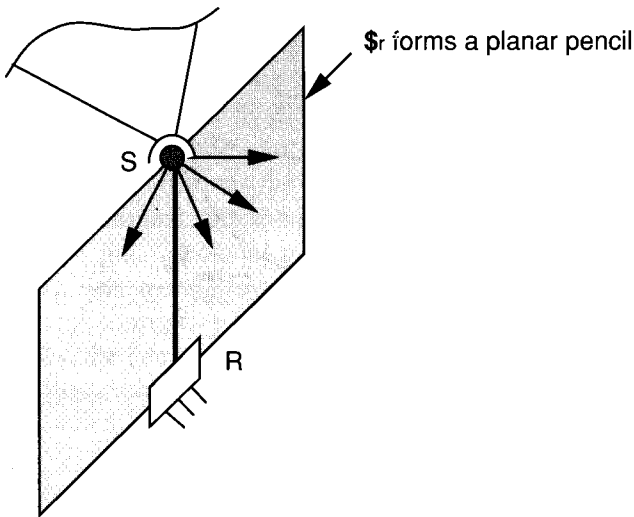


FIGURE 5.10. Revolute–spherical dyad and its reciprocal screw system.

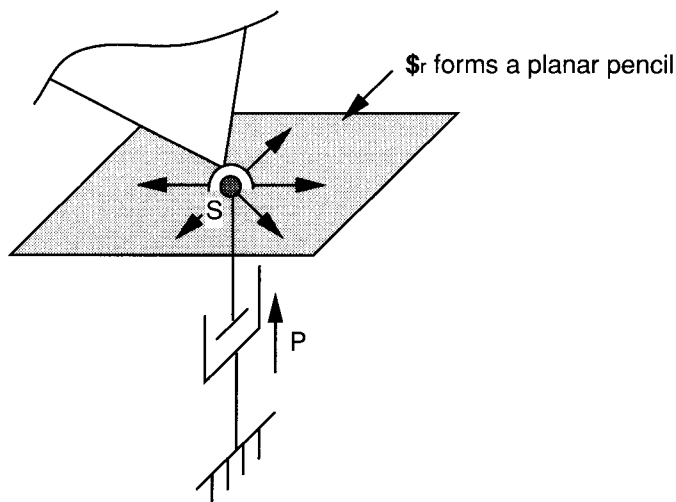


FIGURE 5.11. Prismatic-spherical dyad and its reciprocal screw system.

is perpendicular to the axis of the prismatic joint. Hence all the reciprocal screws are zero-pitch screws forming a planar pencil, as shown in Fig. 5.11.

Surface Contact Pair. Figure 5.12 shows a rigid body B that contacts another body A at point O , where the two contacting surfaces are free to roll

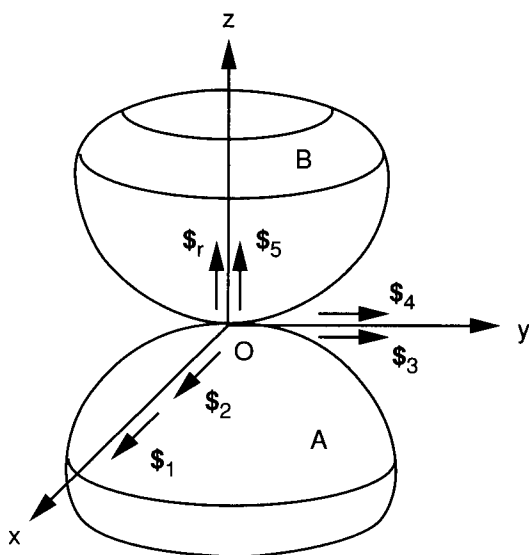


FIGURE 5.12. Two rigid bodies contacting at a point.

and slide with respect to one another. We wish to study the screws associated with this contact pair and its reciprocal screw.

The rigid body B possesses 5 degrees of freedom. Its instantaneous motion can be described by a linear combination of the following five unit screws:

$$\begin{aligned}\hat{\$}_1 &= [1, 0, 0, 0, 0, 0]^T, \\ \hat{\$}_2 &= [0, 0, 0, 1, 0, 0]^T, \\ \hat{\$}_3 &= [0, 1, 0, 0, 0, 0]^T, \\ \hat{\$}_4 &= [0, 0, 0, 0, 1, 0]^T, \\ \hat{\$}_5 &= [0, 0, 1, 0, 0, 0]^T.\end{aligned}\tag{5.43}$$

Since the screws associated with the contact pair form a 5-system, there exists a unique reciprocal screw. Because $\hat{\$}_1$, $\hat{\$}_3$, and $\hat{\$}_5$ are three intersecting zero-pitch screws, the reciprocal screw is a zero-pitch screw passing through the point of contact. Furthermore, due to the presence of two infinite-pitch screws, $\hat{\$}_2$ and $\hat{\$}_4$, the reciprocal screw must be perpendicular to both x and y axes. Hence we conclude that the reciprocal screw is given by

$$\hat{\$}_r = [0, 0, 0, 0, 0, 1]^T.\tag{5.44}$$

5.6 SCREW-BASED JACOBIAN

A parallel manipulator typically consists several limbs, say m . Each limb is made up of several links connected by various joints. Usually, one or more joints in each limb are driven by actuators, while the remaining joints are passive. Figure 5.13 shows the joint screws associated with a typical limb, where the first subscript denotes the joint number and the second subscript represents the limb number. Since we can always replace a cylindrical joint by a revolute joint plus a coaxial prismatic joint, and a spherical joint by three noncoplanar intersecting revolute joints, we may consider each limb as an open-loop chain connecting a moving platform to a fixed base by ℓ 1-dof joints. In this way, the instantaneous twist, $\$_p$, of the moving platform can be expressed as a linear combination of ℓ instantaneous twists (Mohamed and Duffy, 1985):

$$\$ _p = \sum_{j=1}^{\ell} \dot{q}_{j,i} \hat{\$}_{j,i} \quad \text{for } i = 1, 2, \dots, m,\tag{5.45}$$

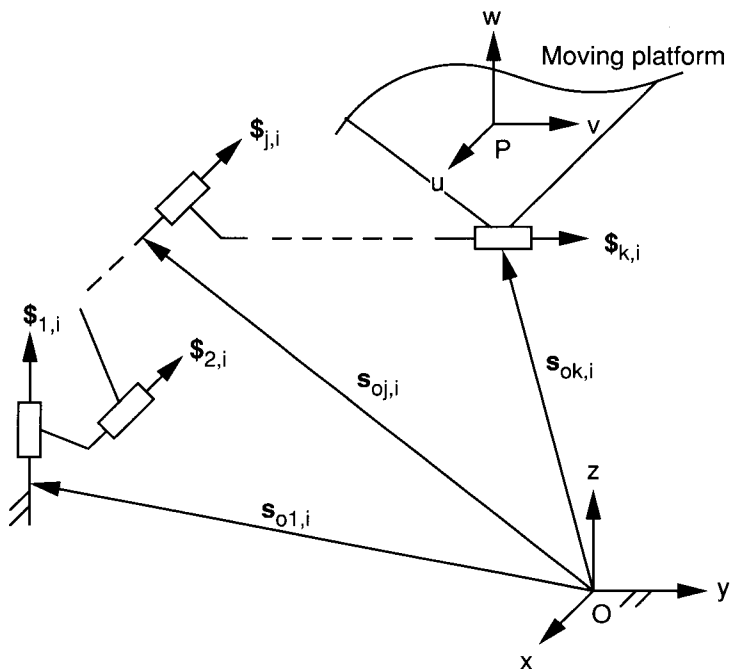


FIGURE 5.13. The i th limb of a parallel manipulator.

where $\dot{q}_{j,i}$ denotes the intensity and $\hat{\$}_{j,i}$ represents a unit screw associated with the j th joint of the i th limb. Recall that the end-effector twist is defined as

$$\$_p = \begin{bmatrix} \omega_n \\ \mathbf{v}_o \end{bmatrix}. \quad (5.46)$$

Equation (5.45) contains many unactuated joint screws which can be eliminated by applying the theory of reciprocal screws. Assuming that the actuated joints in each limb appear in the first g terms, we first identify g unit screws, $\hat{\$}_{r,j,i}$ for $j = 1, 2, \dots, g$, each of which is reciprocal to all the unactuated joint screws in the i th limb. Then we perform the orthogonal product of both sides of Eq. (5.45) with each reciprocal screw. This produces g equations, which can be written in matrix form as

$$J_{x,i} \$_p = J_{q,i} \dot{\mathbf{q}}_i, \quad (5.47)$$

where

$$J_{x,i} = \begin{bmatrix} \hat{\$}_{r1,i}^T \\ \hat{\$}_{r2,i}^T \\ \vdots \\ \hat{\$}_{rg,i}^T \end{bmatrix},$$

$$J_{q,i} = \begin{bmatrix} \hat{\$}_{r1,i}^T \hat{\$}_{1,i} & \hat{\$}_{r1,i}^T \hat{\$}_{2,i} & \cdots & \hat{\$}_{r1,i}^T \hat{\$}_{g,i} \\ \hat{\$}_{r2,i}^T \hat{\$}_{1,i} & \hat{\$}_{r2,i}^T \hat{\$}_{2,i} & \cdots & \hat{\$}_{r2,i}^T \hat{\$}_{g,i} \\ \vdots & \vdots & \cdots & \vdots \\ \hat{\$}_{rg,i}^T \hat{\$}_{1,i} & \hat{\$}_{rg,i}^T \hat{\$}_{2,i} & \cdots & \hat{\$}_{rg,i}^T \hat{\$}_{g,i} \end{bmatrix},$$

$$\dot{\mathbf{q}}_i = [\dot{q}_{1,i}, \dots, \dot{q}_{g,i}]^T.$$

Equation (5.47) written m times, once for each limb, yields $n = m \times g$ linear equations, which can be assembled in the form

$$J_x \$_p = J_q \dot{\mathbf{q}}, \quad (5.48)$$

where

$$J_x = \begin{bmatrix} J_{x,1} \\ J_{x,2} \\ \vdots \\ J_{x,m} \end{bmatrix},$$

$$J_q = \begin{bmatrix} [J_{q,1}] & 0 & \cdots & 0 \\ 0 & [J_{q,2}] & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & [J_{q,m}] \end{bmatrix},$$

$$\dot{\mathbf{q}} = [\dot{q}_{1,1}, \dots, \dot{q}_{g,1}, \dot{q}_{1,2}, \dots, \dot{q}_{g,2}, \dots, \dot{q}_{g,m}]^T.$$

Note that each row of J_x represents the transpose of a reciprocal screw.

The twists associated with all the joints of a limb form an ℓ -system provided that they are linearly independent. Let d be the dimension of the motion space ($d = 6$ for spatial and $d = 3$ for planar and spherical motions). Obviously, if $\ell = d$, there exists no screw reciprocal to the ℓ -system of twists. If $\ell < d$, there are $(d - \ell)$ linearly independent screws which form a $(d - \ell)$ -system. Every screw in the $(d - \ell)$ -system is reciprocal to the ℓ -system of twists. For a limb with k unactuated joints, there exist $(d - k)$ linearly inde-

pendent screws that form a $(d - k)$ -system. Every screw in the $(d - k)$ -system is reciprocal to every unactuated joint screw of the limb. Clearly, the $(d - k)$ -system contains the $(d - \ell)$ -system. Any screw taken from the $(d - k)$ -system, provided that it does not belong to the $(d - \ell)$ -system, can be used as a reciprocal screw, $\hat{\$}_{rj,i}$, to formulate of the Jacobian matrices. Hence there are plenty of reciprocal screws from which to choose. Furthermore, if each reciprocal screw is chosen to be reciprocal to all the joint screws, except for just one of the actuated joint screws, $J_{q,i}$ reduces to

$$J_{q,i} = \begin{bmatrix} \hat{\$}_{r1,1}^T \hat{\$}_{1,1} & 0 & \cdots & 0 \\ 0 & \hat{\$}_{r2,1}^T \hat{\$}_{2,1} & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & \hat{\$}_{rg,1}^T \hat{\$}_{g,1} \end{bmatrix}. \quad (5.49)$$

Hence J_q becomes a diagonal matrix, and the nonzero elements are given by the orthogonal products of the actuated joint screws with their corresponding reciprocal screws.

5.6.1 Screw-based Jacobian of a Stewart–Gough Platform

In this example, the Jacobian matrix of the Stewart–Gough platform shown in Fig. 3.15 is derived by applying the concept of reciprocal screws. Figure 5.14 depicts the equivalent kinematic chain of an *UPS* limb, where the lower spherical joint is replaced by two intersecting unit screws, $\hat{\$}_1$ and $\hat{\$}_2$, and the upper spherical joint is replaced by three intersecting unit screws, $\hat{\$}_4$, $\hat{\$}_5$, and $\hat{\$}_6$, respectively. The prismatic joint is denoted by $\hat{\$}_3$. Thus the passive degree of freedom between the S – S pair is eliminated.

As described in Chapter 3, two Cartesian coordinate systems $A(x, y, z)$ and $B(u, v, w)$ are attached to the fixed base and moving platform, respectively. Furthermore, we define an instantaneous reference frame $C(x', y', z')$, with its origin located at point P and the x' , y' , and z' axes parallel to the x , y , and z axes of the fixed frame. Then we express all the joint screws with respect to this instantaneous reference frame.

There are six joint screws associated with each limb. The third joint is the only actuated joint; the remaining five are passive. The actuated joint screw is an infinite pitch screw. Let $\mathbf{s}_{j,i}$ be a unit vector along the j th joint axis of the i th limb. Then the six unit joint screws of a limb can be written as

$$\hat{\$}_{1,i} = \begin{bmatrix} \mathbf{s}_{1,i} \\ (\mathbf{b}_i - \mathbf{d}_i) \times \mathbf{s}_{1,i} \end{bmatrix},$$

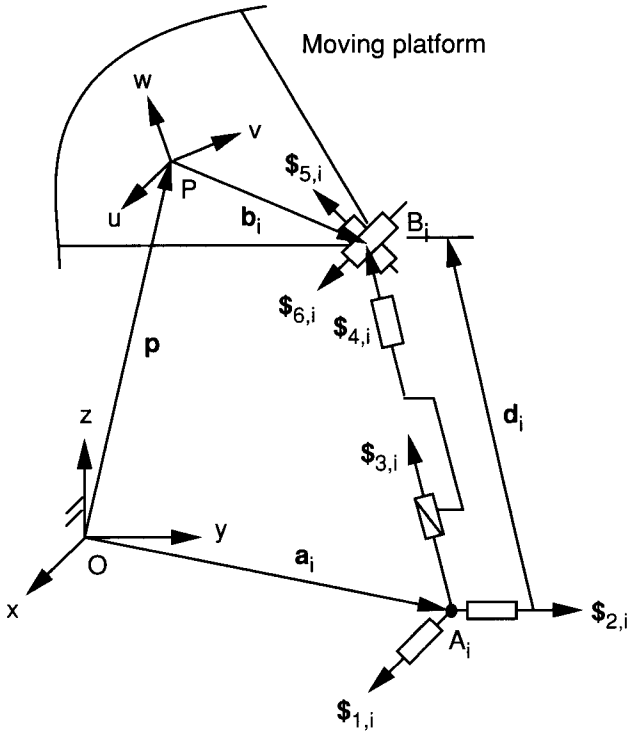


FIGURE 5.14. Equivalent kinematic structure of an *SPS* limb.

$$\hat{\$}_{2,i} = \begin{bmatrix} s_{2,i} \\ (\mathbf{b}_i - \mathbf{d}_i) \times s_{2,i} \end{bmatrix},$$

$$\hat{\$}_{3,i} = \begin{bmatrix} \mathbf{0} \\ s_{3,i} \end{bmatrix},$$

$$\hat{\$}_{4,i} = \begin{bmatrix} s_{4,i} \\ \mathbf{b}_i \times s_{4,i} \end{bmatrix},$$

$$\hat{\$}_{5,i} = \begin{bmatrix} s_{5,i} \\ \mathbf{b}_i \times s_{5,i} \end{bmatrix},$$

$$\hat{\$}_{6,i} = \begin{bmatrix} s_{6,i} \\ \mathbf{b}_i \times s_{6,i} \end{bmatrix},$$

where $\mathbf{b}_i = \overline{PB_i}$, $\mathbf{d}_i = \overline{A_iB_i} = d_i s_{3,i}$, and $s_{3,i} = s_{4,i}$.

We now consider each limb as an open-loop chain and express the instantaneous twist of the moving platform in terms of the joint screws:

$$\$_p = \dot{\theta}_{1,i} \hat{\$_{1,i}} + \dot{\theta}_{2,i} \hat{\$_{2,i}} + \dot{d}_i \hat{\$_{3,i}} + \dot{\theta}_{4,i} \hat{\$_{4,i}} + \dot{\theta}_{5,i} \hat{\$_{5,i}} + \dot{\theta}_{6,i} \hat{\$_{6,i}}. \quad (5.50)$$

Since the axes of all the unactuated joints in each limb intersect the line passing through points A_i and B_i , a unique screw that is reciprocal to all the unactuated joint screws is readily identified as

$$\hat{\$_{r3,i}} = \begin{bmatrix} \mathbf{s}_{3,i} \\ \mathbf{b}_i \times \mathbf{s}_{3,i} \end{bmatrix}. \quad (5.51)$$

Taking the orthogonal product of both sides of Eq. (5.50) with (5.51), we obtain

$$\hat{\$_{r3,i}}^T \$_p = \dot{d}_i \quad \text{for } i = 1, 2, \dots, 6. \quad (5.52)$$

Writing Eq. (5.52) six times, once for each limb, we obtain

$$J_x \dot{\mathbf{x}} = J_q \dot{\mathbf{q}}, \quad (5.53)$$

where

$$J_x = \begin{bmatrix} (\mathbf{b}_1 \times \mathbf{s}_{3,1})^T & \mathbf{s}_{3,1}^T \\ (\mathbf{b}_2 \times \mathbf{s}_{3,2})^T & \mathbf{s}_{3,2}^T \\ \vdots & \vdots \\ (\mathbf{b}_6 \times \mathbf{s}_{3,6})^T & \mathbf{s}_{3,6}^T \end{bmatrix},$$

$$J_q = I(6 \times 6 \text{ identity matrix}),$$

$$\dot{\mathbf{x}} = [\omega_x, \omega_y, \omega_z, v_{px}, v_{py}, v_{pz}]^T,$$

$$\dot{\mathbf{q}} = [\dot{d}_1, \dot{d}_2, \dot{d}_3, \dot{d}_4, \dot{d}_5, \dot{d}_6]^T.$$

We note that $\mathbf{s}_{3,i}$ represents the same unit vector, $\mathbf{s}_i$, used in Eqs. (5.29)–(5.31). Hence when the first and the last three columns of J_x are interchanged, the screw-based Jacobian becomes the conventional Jacobian. We have arrived at the same result as that obtained earlier using the conventional velocity vector-loop method.

5.6.2 Screw-based Jacobian of a Platform Manipulator Driven by Three Planar Motors

Figure 5.15 shows a 6-dof parallel manipulator developed by Tsai and Tahmasebi (1993). This manipulator is made up of a moving platform, a fixed

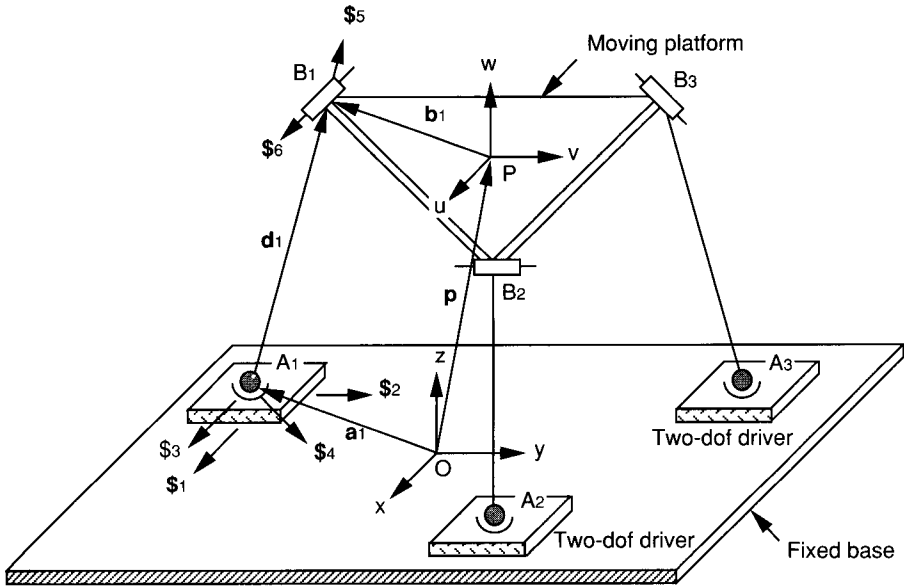


FIGURE 5.15. Platform manipulator driven by three planar motors.

base, and three inextensible limbs, $A_i B_i$ for $i = 1$ to 3. The upper end of each limb is connected to the moving platform by a revolute joint, while the lower end is connected to a planar 2-dof driver by a spherical joint. The revolute joint axes at B_1 , B_2 , and B_3 lie on a plane defined by B_1 , B_2 , and B_3 . The manipulator motion is obtained by moving the planar motors on the x - y plane of the fixed base. The direct kinematics of this manipulator can be found in Tahmasebi and Tsai (1994). In this section, we derive the Jacobian matrix of the manipulator by means of reciprocal screws.

As shown in Fig. 5.15, two Cartesian coordinate systems, $A(x, y, z)$ and $B(u, v, w)$, are attached to the fixed base and moving platform, respectively. The origin of the (u, v, w) frame is located at the centroid P of the moving platform with the u and v axes lying on the plane of $B_1 B_2 B_3$. The location of the moving platform relative to the fixed base is described by a rotation matrix ${}^A R_B$, and the position vector of the centroid $\mathbf{p}$.

Figure 5.15 also shows an equivalent limb in which the spherical joint is replaced by three noncoplanar intersecting revolute joints, $\hat{\$}_{3,i}$, $\hat{\$}_{4,i}$, and $\hat{\$}_{5,i}$. The axis of $\hat{\$}_{3,i}$ is aligned with the x -axis, the axis of $\hat{\$}_{5,i}$ is aligned with the longitudinal axis of the limb, and the axis of $\hat{\$}_{4,i}$ is perpendicular to both $\hat{\$}_{3,i}$ and $\hat{\$}_{5,i}$. The first joint screw, $\hat{\$}_{1,i}$, points along the x -direction. The second joint screw, $\hat{\$}_{2,i}$, points along the y -direction. The sixth joint screw, $\hat{\$}_{6,i}$, lies on the u - v plane. Overall, there are six joint screws associated with the

equivalent limb. The first two are actuated prismatic joints, while the remaining four are passive revolute joints.

To facilitate the analysis, we define an instantaneous reference frame, $C(x', y', z')$, with its origin located at point P and the x' , y' , and z' axes parallel to the x , y , and z axes of the fixed frame. Then we express all the unit joint screws with respect to this instantaneous reference frame as follows:

$$\begin{aligned}\hat{\$}_{1,i} &= \begin{bmatrix} \mathbf{0} \\ \mathbf{s}_{1,i} \end{bmatrix}, \\ \hat{\$}_{2,i} &= \begin{bmatrix} \mathbf{0} \\ \mathbf{s}_{2,i} \end{bmatrix}, \\ \hat{\$}_{3,i} &= \begin{bmatrix} \mathbf{s}_{3,i} \\ (\mathbf{b}_i - \mathbf{d}_i) \times \mathbf{s}_{3,i} \end{bmatrix}, \\ \hat{\$}_{4,i} &= \begin{bmatrix} \mathbf{s}_{4,i} \\ (\mathbf{b}_i - \mathbf{d}_i) \times \mathbf{s}_{4,i} \end{bmatrix}, \\ \hat{\$}_{5,i} &= \begin{bmatrix} \mathbf{s}_{5,i} \\ \mathbf{b}_i \times \mathbf{s}_{5,i} \end{bmatrix}, \\ \hat{\$}_{6,i} &= \begin{bmatrix} \mathbf{s}_{6,i} \\ \mathbf{b}_i \times \mathbf{s}_{6,i} \end{bmatrix},\end{aligned}$$

where $\mathbf{s}_{1,i} = \mathbf{s}_{3,i} = [1, 0, 0]^T$, $\mathbf{s}_{2,i} = [0, 1, 0]^T$, $\mathbf{a}_i = \overline{OA_i}$, $\mathbf{b}_i = \overline{PB_i}$, $\mathbf{d}_i = \overline{A_iB_i}$, $\mathbf{s}_{5,i} = \mathbf{d}_i/d_i$, and $\mathbf{s}_{4,i} = \mathbf{s}_{5,i} \times \mathbf{s}_{3,i}$.

We now consider each limb as an open-loop chain and express the instantaneous twist, $\$p$, of the moving platform in terms of the joint screws:

$$\$p = v_{x,i} \hat{\$}_{1,i} + v_{y,i} \hat{\$}_{2,i} + \dot{\theta}_{3,i} \hat{\$}_{3,i} + \cdots + \dot{\theta}_{6,i} \hat{\$}_{6,i} \quad i = 1, 2, 3, \quad (5.54)$$

where $v_{x,i}$ and $v_{y,i}$ denote the linear velocities of the planar motor along the x and y directions, and $\dot{\theta}_{3,i}, \dots, \dot{\theta}_{6,i}$ denote the passive joint rates.

Since there is a spherical joint at the lower end and a revolute joint at the upper end, all screws that are reciprocal to the unactuated joint screws form a planar pencil. Specifically, all the reciprocal screws are zero-pitch screws passing through A_i and lying on a plane H_i that contains the axis of $\hat{\$}_{6,i}$ and point A_i .

Theoretically, any two screws taken from the reciprocal screw system contained in H_i can be used to formulate the Jacobian matrix. Perhaps the most straightforward approach is to identify two reciprocal screws, each reciprocal to all the joint screws except for only one of the two infinite pitch screws, $\hat{\$}_{1,i}$ and $\hat{\$}_{2,i}$.

Let $\mathbf{n}_i = [n_{x,i}, n_{y,i}, n_{z,i}]^T$ be a unit vector defined by the cross product of $\mathbf{s}_{5,i}$ and $\mathbf{s}_{6,i}$; that is,

$$\mathbf{n}_i = \mathbf{s}_{5,i} \times \mathbf{s}_{6,i}. \quad (5.55)$$

Then a screw that is reciprocal to all screws except for $\hat{\$}_{1,i}$ is obtained by an intersection of the H_i -plane and a plane that passes point A_i and is parallel to the x - z plane.

$$\hat{\$}_{r1,i} = \left[\begin{array}{c} \mathbf{s}_{r1,i} \\ (\mathbf{b}_i - \mathbf{d}_i) \times \mathbf{s}_{r1,i} \end{array} \right], \quad \text{where} \quad \mathbf{s}_{r1,i} = \frac{[n_{z,i} \ 0 \ -n_{x,i}]^T}{\sqrt{1 - n_{y,i}^2}}. \quad (5.56)$$

Similarly, a screw that is reciprocal to all screws except for $\hat{\$}_{2,i}$ is obtained by an intersection of the H_i -plane and a plane that passes point A_i and is parallel to the y - z plane.

$$\hat{\$}_{r2,i} = \left[\begin{array}{c} \mathbf{s}_{r2,i} \\ (\mathbf{b}_i - \mathbf{d}_i) \times \mathbf{s}_{r2,i} \end{array} \right], \quad \text{where} \quad \mathbf{s}_{r2,i} = \frac{[0 \ n_{z,i} \ -n_{y,i}]^T}{\sqrt{1 - n_{x,i}^2}}. \quad (5.57)$$

Taking the orthogonal products of both sides of Eq. (5.54) with (5.56) and (5.57) in turn, the resulting equations can be arranged in matrix form:

$$\left[\begin{array}{cc} ((\mathbf{b}_i - \mathbf{d}_i) \times \mathbf{s}_{r1,i})^T & \mathbf{s}_{r1,i}^T \\ ((\mathbf{b}_i - \mathbf{d}_i) \times \mathbf{s}_{r2,i})^T & \mathbf{s}_{r2,i}^T \end{array} \right] \left[\begin{array}{c} \boldsymbol{\omega}_p \\ \mathbf{v}_p \end{array} \right] = \left[\begin{array}{cc} \frac{n_{z,i}}{\sqrt{1 - n_{y,i}^2}} & 0 \\ 0 & \frac{n_{z,i}}{\sqrt{1 - n_{x,i}^2}} \end{array} \right] \left[\begin{array}{c} v_{x,i} \\ v_{y,i} \end{array} \right]. \quad (5.58)$$

Writing Eqs. (5.58) three times, once for each limb, we obtain

$$J_x \dot{\mathbf{x}} = J_q \dot{\mathbf{q}}, \quad (5.59)$$

where

$$J_x = \left[\begin{array}{cc} ((\mathbf{b}_1 - \mathbf{d}_1) \times \mathbf{s}_{r1,1})^T & \mathbf{s}_{r1,1}^T \\ ((\mathbf{b}_1 - \mathbf{d}_1) \times \mathbf{s}_{r2,1})^T & \mathbf{s}_{r2,1}^T \\ ((\mathbf{b}_2 - \mathbf{d}_2) \times \mathbf{s}_{r1,2})^T & \mathbf{s}_{r1,2}^T \\ ((\mathbf{b}_2 - \mathbf{d}_2) \times \mathbf{s}_{r2,2})^T & \mathbf{s}_{r2,2}^T \\ ((\mathbf{b}_3 - \mathbf{d}_3) \times \mathbf{s}_{r1,3})^T & \mathbf{s}_{r1,3}^T \\ ((\mathbf{b}_3 - \mathbf{d}_3) \times \mathbf{s}_{r2,3})^T & \mathbf{s}_{r2,3}^T \end{array} \right],$$

$$J_q = \begin{bmatrix} \frac{n_{z,1}}{\sqrt{1-n_{y,1}^2}} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{n_{z,1}}{\sqrt{1-n_{x,1}^2}} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{n_{z,2}}{\sqrt{1-n_{y,2}^2}} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{n_{z,2}}{\sqrt{1-n_{x,2}^2}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{n_{z,3}}{\sqrt{1-n_{y,3}^2}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{n_{z,3}}{\sqrt{1-n_{x,3}^2}} \end{bmatrix},$$

$$\dot{\mathbf{x}} = [\omega_x, \omega_y, \omega_z, v_{px}, v_{py}, v_{pz}]^T,$$

$$\dot{\mathbf{q}} = [v_{x,1}, v_{y,1}, v_{x,2}, v_{y,2}, v_{x,3}, v_{y,3}]^T.$$

Although it is difficult to identify all the possible singular configurations of this manipulator, some of them can be found by inspection of the J_q and J_x matrices.

(a) Inverse Kinematic Singularities. We note that the nonzero elements of J_q are proportional to the z -components of the three unit vectors $\mathbf{n}_i$ for $i = 1, 2$, and 3 . Hence an inverse kinematic singularity occurs when $n_{z,i} = 0$ for $i = 1$ or 2 or 3 . Physically, when one of the limbs points in the z -direction, the manipulator loses 2 degrees of freedom. Under such a condition, a differential motion of the planar motor about the x or y axis does not affect the velocity of the moving platform. If two or three limbs point in the z -direction simultaneously, 4 or 6 degrees of freedom will be lost.

(b) Direct Kinematic Singularities. A trivial direct kinematic singularity occurs when $n_{x,i} = n_{y,i} = 0$ for all the limbs. Under such a condition, the sixth column of J_x becomes zero identically, and the manipulator gains 1 degree of freedom. Physically, this corresponds to the configuration in which both the moving platform and the three limbs lie on the x - y plane. Hence the moving platform can make an infinitesimal translation along the z -axis while all the actuators are locked.

A second direct kinematic singularity occurs when the following three conditions are satisfied simultaneously: (1) $\triangle B_1 B_2 B_3$ is an equilateral triangle; (2) the three limbs are of equal length; and (3) A_1, A_2 , and A_3 are placed directly under the centroid of the moving platform. When these conditions are

$$J_q = \begin{bmatrix} \frac{n_{z,1}}{\sqrt{1-n_{y,1}^2}} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{n_{z,1}}{\sqrt{1-n_{x,1}^2}} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{n_{z,2}}{\sqrt{1-n_{y,2}^2}} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{n_{z,2}}{\sqrt{1-n_{x,2}^2}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{n_{z,3}}{\sqrt{1-n_{y,3}^2}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{n_{z,3}}{\sqrt{1-n_{x,3}^2}} \end{bmatrix},$$

$$\dot{\mathbf{x}} = [\omega_x, \omega_y, \omega_z, v_{px}, v_{py}, v_{pz}]^T,$$

$$\dot{\mathbf{q}} = [v_{x,1}, v_{y,1}, v_{x,2}, v_{y,2}, v_{x,3}, v_{y,3}]^T.$$

Although it is difficult to identify all the possible singular configurations of this manipulator, some of them can be found by inspection of the J_q and J_x matrices.

(a) Inverse Kinematic Singularities. We note that the nonzero elements of J_q are proportional to the z -components of the three unit vectors $\mathbf{n}_i$ for $i = 1, 2$, and 3 . Hence an inverse kinematic singularity occurs when $n_{z,i} = 0$ for $i = 1$ or 2 or 3 . Physically, when one of the limbs points in the z -direction, the manipulator loses 2 degrees of freedom. Under such a condition, a differential motion of the planar motor about the x or y axis does not affect the velocity of the moving platform. If two or three limbs point in the z -direction simultaneously, 4 or 6 degrees of freedom will be lost.

(b) Direct Kinematic Singularities. A trivial direct kinematic singularity occurs when $n_{x,i} = n_{y,i} = 0$ for all the limbs. Under such a condition, the sixth column of J_x becomes zero identically, and the manipulator gains 1 degree of freedom. Physically, this corresponds to the configuration in which both the moving platform and the three limbs lie on the x - y plane. Hence the moving platform can make an infinitesimal translation along the z -axis while all the actuators are locked.

A second direct kinematic singularity occurs when the following three conditions are satisfied simultaneously: (1) $\triangle B_1 B_2 B_3$ is an equilateral triangle; (2) the three limbs are of equal length; and (3) A_1, A_2 , and A_3 are placed directly under the centroid of the moving platform. When these conditions are

met, the third column of J_x goes to zero, the first and fifth columns become linearly dependent, and the second and fourth columns also become linearly dependent. Hence the manipulator gains 3 rotational degrees of freedom.

A third direct kinematic singularity occurs when $n_{z,i} = 0$ for $i = 1, 2$, and 3. When the three limbs are all pointing along the z -direction, the fourth and fifth columns of J_x are both equal to zero. The moving platform can make an infinitesimal translation along any direction parallel to the x - y plane. Hence the manipulator gains 2 degrees of freedom. In fact, this is a combined singularity, since under such a condition the manipulator is also under an inverse kinematic singularity.

5.7 SUMMARY

In this chapter, the Jacobian and singular conditions of parallel manipulators have been studied. Both the conventional and screw-based Jacobian matrices were discussed. The velocity vector-loop method was employed for the derivation of conventional Jacobians, while the concept of reciprocal screws was used for the derivation of screw-based Jacobians. It was shown that the Jacobian of a parallel manipulator can conveniently be split into two matrices: one associated with the direct kinematics and the other with the inverse kinematics. Depending on which matrix is singular, the singularities of a parallel manipulator can be classified into three types. At an inverse kinematic singular configuration, a manipulator loses 1 or more degrees of freedom, while at a direct kinematic singular configuration, a manipulator gains 1 or more degrees of freedom. A combined singularity results when the inverse and direct kinematic singularities occur simultaneously. Generally, a combined singularity depends not only on the manipulator configuration but also on the kinematic architecture. The classification and physical significance of each type of singularity were illustrated by several examples.

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EXERCISES

1. Calculate the end-effector velocity $\mathbf{v}_q$ as a function of input joint angles and joint rates for the planar $5R$ manipulator shown in Fig. 1.7. Sketch the singular loci of the manipulator.
2. Derive the Jacobian matrix of the planar $4R1P$ manipulator shown in Fig. 3.19. Identify the singular configurations.
3. We wish to move the moving platform of the planar $3PRP$ manipulator shown in Fig. 3.3 along the x -axis with a linear velocity of $\mathbf{v}_q$ m/s and zero angular velocity. Calculate the linear velocities of the input actuators.
4. Derive the conventional Jacobian matrix of the DELTA robot shown in Fig. 3.21 by using the velocity vector-loop approach. Under what conditions will the manipulator gain extra degree(s) of freedom?
5. Derive the conventional Jacobian matrix of the 3×3 Stewart platform shown in Fig. 3.16 by the velocity vector-loop approach. Does this manipulator possess any inverse kinematics singularities?
6. Find the reciprocal screws of a spherical–universal dyad assuming that the axes of the universal joints do not pass through the center of the sphere.
7. Find the reciprocal screws of a spherical–cylindrical dyad assuming that the axis of the cylindrical joint does not pass through the center of the sphere.
8. Formulate the Jacobian matrix of the 3×3 Stewart platform shown in Fig. 3.16 by the method of reciprocal screws.